

Deep Learning for Automated Brain Tumor Segmentation

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Abstract:

This paper presents a comprehensive comparison of various deep learning architectures for automated brain tumor segmentation. We evaluated UNet3D, VNet, DeepMedic, and our novel BrainTumorSegModel on the BraTS dataset. Results demonstrate that our proposed architecture achieves state-of-the-art performance with a Dice score of 0.92 for whole tumor segmentation, while maintaining efficient computational requirements.

Keywords: brain tumor, segmentation, deep learning, MRI, neural networks

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